ROOT LOCUS ON THE Z-PLANE: DISCRETE P DESIGN FOR ANGLE CONTROL

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Discretise the angle model of the DC motor with a zero-order hold and identify its poles and zero in the z-plane.	☐
2	Use the z-plane root locus to choose a proportional gain that meets an overshoot requirement at a given sample period.	☐
3	Compare the z-plane design with the continuous-time design on the hardware, and explain when a continuous-time design can safely be reused.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The angle of the shaft is measured from the encoder counts, as described in the practical exercise, Manual PID Tuning of DC Motor Angle Control, and the output voltage of the controller sets both the direction and the PWM value. Because the angle is the integral of the speed, the angle model is $G_\theta(s) = K_\theta/(s(\tau s + 1))$, where $K_\theta = 6K$ converts RPM/V into degrees per second per volt. For the A1.4 example motor, $K_\theta = 168$ °/(s·V).

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The proportional angle controller of the practical exercise, Root Locus: Designing a Gain for a Damping Target, is redesigned directly in the z-plane, including the zero-order hold. The two designs are compared at a fast and a deliberately slow

sample period. The exercise supports the ACE-CORE learning guide, Proportional Control Design using Root Locus on the z-Plane.

Discretise the Angle Model

With a zero-order hold at sample period T_s, the angle model becomes a second-order discrete transfer function with poles at $z = 1$ (the integrator) and $z = a = e^{-T_s/\tau}$, and one zero introduced by the hold:

$$G_\theta(z) = \mathcal{Z}\Big\{\frac{1 - e^{-sT_s}}{s} \cdot \frac{K_\theta}{s(\tau s + 1)}\Big\} = \frac{b_1 z + b_0}{(z - 1)(z - a)}$$

The root locus construction rules apply in the z-plane as in the s-plane, but the stability boundary is the unit circle and lines of constant ζ become spirals, given by $z = e^{sT}$ for s on the ζ line. The gain is chosen where the locus crosses the $\zeta = 0.7$ contour.

The z-plane root locus for the example motor at $T_s = 0.05$ s is shown in Figure 1. The locus leaves the real axis between the two poles and curves round the ZOH zero on the negative real axis. It crosses the $\zeta = 0.7$ contour at $K_p = 0.0173$ V/°, lower than the continuous-time design of 0.0202 V/°.

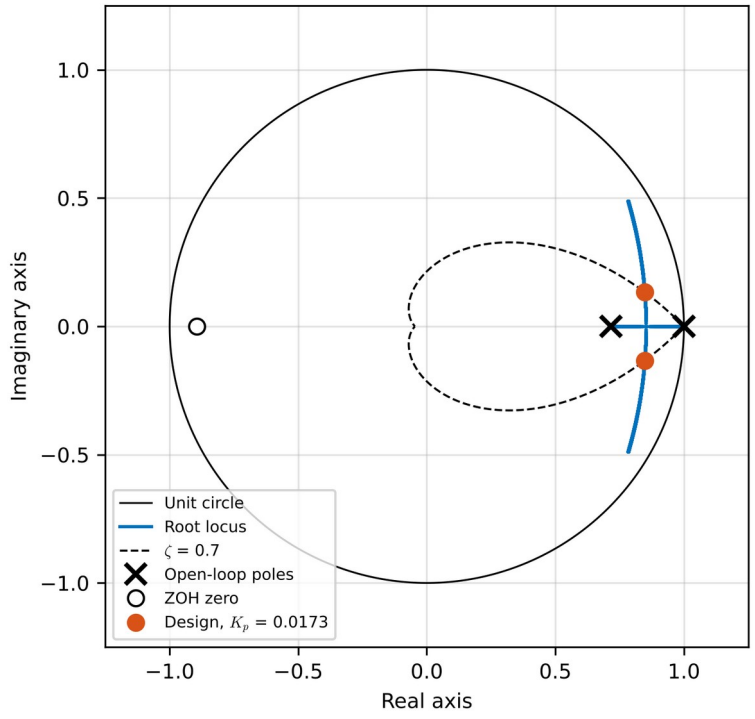


Figure 1: *Root Locus in the z-Plane for the A1.4 Example Motor at Ts = 0.05 s, Showing the ζ = 0.7 Contour and Design Point*

MATLAB Script: designing the proportional gain in the z-plane

```
Ktheta = 168;  tau = 0.15;          % your angle model [deg/(s V), s]
Ts = 0.05;                          % sample period [s]
Gz = c2d(tf(Ktheta, [tau 1 0]), Ts, 'zoh')   % discrete angle model
rlocus(Gz); zgrid(0.7, []);          % locus and the zeta = 0.7 contour
axis equal; axis([-1.2 1.2 -1.2 1.2]);
% Click on the locus where it crosses the zeta = 0.7 contour to read Kp,
% then check the closed loop:
Kp = 0.0173;
step(90*feedback(Kp*Gz, 1)); grid on;
```

Compare the Designs on the Hardware

1. Set the angle control model to T_s = 0.0075 s (about $\tau/20$). Find the z-plane design gain for this sample period and compare it with the continuous design from the practical exercise, Root Locus: Designing a Gain for a Damping Target.
2. Apply a 90° step with each gain and measure the overshoot.
3. Repeat at T_s = 0.05 s (about $\tau/3$), where the two gains differ more.
4. Record the results in Table 2.

Table 2: *Comparison of z-Plane and Continuous-Time Designs*

T_s (s)	z-plane K_p (V/°)	Sim. OS (%)	Continuous K_p (V/°)	Sim. OS (%)	Measured OS, z-plane (%)	Measured OS, continuous (%)
0.0075	0.0197	4.6	0.0202	5.0		
0.05	0.0173	4.6	0.0202	7.4		

The simulated values in Table 2 use the exact ZOH model of the example motor. At the fast sample period the two gains are almost identical and give the same response. At the slow sample period, the continuous-time gain gives noticeably more overshoot than intended, while the z-plane design still meets the requirement, because it accounts for the lag added by the hold.

1. Where do the lines of constant ζ from the s-plane end up in the z-plane, and why are they not straight?
2. What role does the ZOH zero play in the shape of the locus?
3. Above roughly what sample rate, relative to τ, would you consider a continuous-time design safe to reuse directly?

4.0 Concluding Remarks

This practical exercise designed a proportional angle controller directly in the z-plane, including the zero-order hold, and compared it with the continuous-time

design. Using MATLAB, Simulink and ACE-Lab hardware, the angle model was discretised, its root locus was drawn with the unit circle and constant-ζ contours, and both designs were tested at a fast and a slow sample period.

At a fast sample period the two designs agreed. At a slow one, the hold added lag that the continuous-time design ignored, giving extra overshoot, whereas the z-plane design still met the requirement. Designing in the z-plane is therefore the reliable approach whenever the sample period is a significant fraction of the time constants being controlled.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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